Escaping Sequences under Theta-Source Orthogonality Constraints

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Research note — 17 October 2026

Abstract

We consider the minimum closed even jump form associated with the infinite theta source, an archimedean translation kernel, and all prime-power shifts. Let $\mathcal{N}$ be the closed space generated by the constant and the centered even derivative ratios of the source. We construct real smooth unit vectors $h_m \in \mathcal{D} \cap \mathcal{N}^\perp$ for which $|\mathcal{E}(h_m) - 1/2| \le Cm^3 e^{-\sqrt{m}}$ and the mass in every fixed bounded interval tends to zero. Every defining orthogonality condition is exact. The construction uses confluent divided differences at clusters of zeros of the completed zeta function and the unconditional zero-counting asymptotic; neither real nor simple zeros are assumed. We derive the source identity from the classical compact-test explicit formula and justify all minimum-domain admissions. For a positive two-shift envelope we also prove a ground-state identity and show that its transformed energy tends to zero on the same sequence. These results obstruct a positive uniform margin and fixed-center coercivity under the known source constraints. They do not determine the critical sign of the form, identify the full nullspace, or give an operator Weyl sequence. No conclusion about the Riemann hypothesis is obtained.

1 The form and the constrained escape statement

For $x \ge 0$ define

$$\kappa(x) = e^{x/2} \sum_{n \ge 1} (4z_n(x)^2 - 6z_n(x)) e^{-z_n(x)}, \qquad z_n(x) = \pi n^2 e^{2x}, \tag{1.1}$$

and extend κ evenly. Set

$$c(x) = \cosh(x/2), \qquad \mathrm{d}\nu(x) = 2\kappa(x)c(x)\,\mathrm{d}x, \qquad \mathcal{H} = L^2_{\mathrm{even}}(\mathbb{R}, \nu; \mathbb{C}). \tag{1.2}$$

The normalization $\nu(\mathbb{R}) = 1$ is proved below. All inner products are conjugate-linear in the first variable. We use $\tau_s u(x) = u(x+s)$ and $\mathrm{Var}_\nu(h) = \|h\|^2_{\mathcal{H}} - |\int h\,\mathrm{d}\nu|^2$. Write

$$\begin{aligned} \rho(s) &= \frac{e^{-s/2}}{1 - e^{-2s}}, \quad s > 0, \qquad c_n = \frac{\Lambda(n)}{\sqrt{n}}, \quad s_n = \log n, \\ \mathrm{d}\mu(s) &= \frac{1}{2}\rho(|s|)\,\mathrm{d}s + \frac{1}{2}\sum_{n \ge 2} c_n(\delta_{s_n} + \delta_{-s_n}). \end{aligned} \tag{1.3}$$

Here Λ includes every prime power. The positive form on $\mathcal{C} = C_c^\infty(\mathbb{R};\mathbb{C}) \cap \mathcal{H}$ is

$$\begin{aligned}\mathcal{E}(h) &= \iint \kappa(x)\kappa(z)|h(x) - h(z)|^2 \,\mathrm{d}\mu(x-z)\,\mathrm{d}x \\ &= \int_0^\infty \rho(s) \int_{\mathbb{R}} \kappa(x)\kappa(x+s)|h(x+s) - h(x)|^2 \,\mathrm{d}x\,\mathrm{d}s \\ &\quad + \sum_{n\geq 2} c_n \int_{\mathbb{R}} \kappa(x)\kappa(x+s_n)|h(x+s_n) - h(x)|^2 \,\mathrm{d}x. \end{aligned} \tag{1.4}$$

Thus the first line uses a symmetric half measure. In a double integral with $\mathrm{d}\mu(x-z)\,\mathrm{d}x$, the variable z is integrated against the translated shift measure; atoms are not replaced by a Lebesgue density.

The form is closable, and $(\mathcal{E}, \mathcal{D})$ denotes its *minimum* closure from $\mathcal{C}$ in the norm

$$\|h\|_{\mathcal{D}}^2 = \mathcal{E}(h) + \|h\|_{\mathcal{H}}^2. \tag{1.5}$$

No equality with a maximal jump-energy domain is assumed. Let $A \geq 0$ be the representing self-adjoint operator and set

$$P_1 h = \langle 1, h\rangle_{\mathcal{H}}\, 1, \qquad L = A - \frac{1}{2}I + \frac{1}{2}P_1, \qquad \mathrm{D}(L) = \mathrm{D}(A). \tag{1.6}$$

The known source-generated space considered here is

$$f_j = \frac{\kappa^{(2j)}}{\kappa} \quad (j \geq 0), \qquad \mathcal{N} = \overline{\mathrm{span}}_{\mathcal{H}}\{f_j : j \geq 0\}. \tag{1.7}$$

We prove $f_j \in \mathcal{H}$ and $\langle 1, f_j\rangle_{\mathcal{H}} = 4^{-j}$ below, so this is also the closed span of 1 and $f_j - 4^{-j}$ for $j \geq 1$.

Theorem 1.1 (Escape under every known source constraint)**.** *There exist constants $C < \infty$, m_0, and real smooth even functions $h_m \in \mathcal{D} \cap \mathcal{N}^\perp$ for all integers $m \geq m_0$ such that*

$$\|h_m\|_{\mathcal{H}} = 1, \qquad |\mathcal{E}(h_m) - \tfrac{1}{2}| \leq Cm^3 e^{-\sqrt{m}}. \tag{1.8}$$

For every fixed $T < \infty$,

$$\|\mathbf{1}_{[-T,T]} h_m\|_{\mathcal{H}} \longrightarrow 0, \qquad h_m \rightharpoonup 0 \text{ in } \mathcal{H}. \tag{1.9}$$

Each moment $\langle f_j, h_m\rangle_{\mathcal{H}}$ is exactly zero for every $j \geq 0$. For each fixed admissible envelope R of Definition 6.2, there are constants $C_R < \infty$ and $a_R > 0$ such that, after increasing the starting index if necessary,

$$0 \leq D_R(h_m) \leq C_R m^3 e^{-a_R\sqrt{m}} \longrightarrow 0. \tag{1.10}$$

Corollary 1.2. *On $\mathcal{D} \cap \mathcal{N}^\perp$ there is no inequality $\mathcal{E}(h) \geq (1/2 + \varepsilon)\|h\|_{\mathcal{H}}^2$ with $\varepsilon > 0$. For fixed finite T there is no $a > 0$ such that $\|\mathbf{1}_{[-T,T]}h\|_{\mathcal{H}} \geq a\|h\|_{\mathcal{H}}$, even among unit vectors with $\mathcal{E}(h) \leq 1/2 + \varepsilon$ for any fixed $\varepsilon > 0$. For an admissible fixed R, no finite nonnegative a, b and fixed T can satisfy*

$$\|h\|_{\mathcal{H}}^2 \leq aD_R(h) + b\|\mathbf{1}_{[-T,T]}h\|_{\mathcal{H}}^2 \quad (h \in \mathcal{D} \cap \mathcal{N}^\perp). \tag{1.11}$$

In particular D_R has no positive uniform lower bound by $\|h\|_{\mathcal{H}}^2$ on this space.

The theorem imposes all conditions from the *known source space* $\mathcal{N}$. Although $\mathcal{N} \subset \ker L$ is proved in Proposition 2.7, we do not prove $\mathcal{N} = \ker L$ and do not describe the result as removal of the full nullspace. The critical inequality $\mathcal{E}(h) \geq \|h\|_{\mathcal{H}}^2/2$ on $\mathcal{D} \cap \mathcal{N}^\perp$ remains open here. No operator-norm Weyl sequence or essential-spectrum conclusion is asserted.

The ingredients are classical theta identities, the compact-test explicit formula, unconditional zero counting, and divided differences. We specify their use rather than make a claim of bibliographical priority; a systematic novelty survey is outside this draft. The constants and starting indices are proved to exist, but are not numerically certified.

2 Theta normalization and the source identity

2.1 The exact theta source

Lemma 2.1. *The function κ is positive, smooth, and even. With $\Xi(z) = \xi(1/2 + iz)$ and $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, one has*

$$F_\kappa(z) := \int_{\mathbb{R}} \kappa(x)e^{-izx}\,\mathrm{d}x = \Xi(z), \qquad \int_{\mathbb{R}} \kappa(x)e^{\pm x/2}\,\mathrm{d}x = \frac{1}{2}. \tag{2.1}$$

For $x \geq 0$ there are fixed positive constants a_0, a_1 such that

$$a_0 e^{9x/2-\pi e^{2x}} \leq \kappa(x) \leq a_1 e^{9x/2-\pi e^{2x}}, \qquad a_0 = 2\pi^2. \tag{2.2}$$

For each fixed integer $r \geq 0$,

$$|\kappa^{(r)}(x)| \leq C_r e^{2r|x|}\kappa(x). \tag{2.3}$$

All fixed derivatives decrease faster than every exponential at both ends.

Proof. Let $S(x) = e^{x/2}\sum_{n\geq 1} e^{-\pi n^2 e^{2x}}$. The Jacobi transformation [4, Eq. 20.7.32] gives

$$S(x) - S(-x) = \frac{1}{2}(e^{-x/2} - e^{x/2}).$$

Since $\kappa = (\partial_x^2 - 1/4)S$, the source is even. The series and each fixed derivative converge locally uniformly. For $x \geq 0$, $z_n \geq \pi > 3$, so every original summand is positive. The Mellin calculation, initially absolutely convergent for $\Re s > 1$, is

$$\begin{aligned}\int_{\mathbb{R}} \kappa(x)e^{(s-1/2)x}\,\mathrm{d}x &= \zeta(s)\int_0^\infty (4\pi^2 y^4 - 6\pi y^2)e^{-\pi y^2} y^{s-1}\,\mathrm{d}y \\ &= \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).\end{aligned} \tag{2.4}$$

The standard completed-zeta identities are recorded in [4, Eqs. 25.4.3–25.4.4]. The left side continues to an entire function by the two-sided decay. Evenness and $\xi(0) = \xi(1) = 1/2$ give (2.1) and $\nu(\mathbb{R}) = 1$.

For the estimates put $y = \pi e^{2x}$. The first summand and $4y^2 - 6y \geq 2y^2$ give the lower bound. The upper bound follows from

$$\sum_{n\geq 1} n^4 e^{-(n^2-1)y} \leq \sum_{n\geq 1} n^4 e^{-3(n^2-1)} < \infty.$$

After r derivatives the polynomial in z_n has degree $r+2$: $P_0(z) = 4z^2 - 6z$ and $P_{r+1} = P_r/2 + 2zP_r' - 2zP_r$. The same estimate with n^{2r+4} proves (2.3); its constant may depend on the fixed order. □

Lemma 2.2 (Closability)**.** *The form (1.4) is densely defined and closable on $\mathcal{C}$.*

Proof. The combined positive continuous and atomic edge measure is sigma-finite after restricting bounded endpoint coordinates and shifts bounded away from zero. Each restricted endpoint marginal is absolutely continuous with respect to Lebesgue measure; countably many atomic translations preserve null sets. If $h_k \to 0$ in $\mathcal{H}$ and their edge differences are Cauchy in edge L^2, choose a subsequence converging Lebesgue almost everywhere, using the strict positivity of the weight. The edge differences then tend to zero at almost every pair of endpoints. Their edge-L^2 limit is zero, which proves closability. Core energies are finite by the small-shift derivative bound and theta decay. Core density in $\mathcal{H}$ follows from the positive locally smooth weight and even compact approximation. □

2.2 Classical explicit formula with the constants fixed

For physical even functions define

$$D_\Gamma(u) = \int_0^\infty \rho(s)\|u - \tau_s u\|_2^2 \,\mathrm{d}s, \qquad \mathfrak{m}(u) = \int_{\mathbb{R}} u(x)c(x)\,\mathrm{d}x,$$
$$Q(u) = D_\Gamma(u) - c_\Gamma\|u\|_2^2 - 2\sum_{n\ge 2} c_n \Re\langle u, \tau_{s_n} u\rangle_2 + 2|\mathfrak{m}(u)|^2,$$
$$c_\Gamma = \log\pi + \gamma_E + \frac{\pi}{2} + 3\log 2. \tag{2.5}$$

Polarization defines $Q(u,v)$. These expressions are not asserted to be nonnegative.

Lemma 2.3 (Classical input in physical coordinates)**.** *For $u \in C_c^\infty(\mathbb{R};\mathbb{C})$ even, let $F_u(z) = \int_{\mathbb{R}} u(x)e^{-izx}\,\mathrm{d}x$. Then*

$$Q(u) = \sum_\rho F_u(z_\rho)\overline{F_u(\overline{z_\rho})}, \qquad z_\rho = \frac{\rho - 1/2}{i}, \tag{2.6}$$

where the sum is over nontrivial zeta zeros with multiplicities. The series converges absolutely. No zero-location hypothesis beyond the classical critical strip is used. With the first variable conjugate-linear, the polarized identity is

$$Q(u,v) = \sum_\rho \overline{F_u(\overline{z_\rho})}F_v(z_\rho) \qquad (u, v \in C_c^\infty(\mathbb{R};\mathbb{C}) \text{ even}). \tag{2.7}$$

Proof. We apply the classical multiplicative explicit formula for $\Phi \in C_c^\infty((0,\infty))$, with Mellin transform $\widetilde{\Phi}(s) = \int_0^\infty \Phi(y)y^{s-1}\,\mathrm{d}y$ and $\Phi^\sharp(y) = y^{-1}\Phi(y^{-1})$. Its normalization, prime terms, and coupled archimedean finite part are recorded in [1, Appendix B, Eqs. (147)–(150)]. Put

$$C_u(s) = \int_{\mathbb{R}} \overline{u(x)}u(x+s)\,\mathrm{d}x, \qquad \Phi(y) = y^{-1/2}C_u(\log y).$$

For an even complex u, C_u is real and even. Thus Φ is a permitted smooth compact multiplicative test. Direct integration gives

$$\widetilde{\Phi}(1/2 + iz) = F_u(z)\overline{F_u(\bar{z})}.$$

The two pole terms are $2|\mathfrak{m}(u)|^2$, and the subtracted prime contribution is $2\sum_{n\ge2} c_n C_u(s_n)$. The negative archimedean term, after $y = e^s$, is

$$-(\log(4\pi) + \gamma_E)\|u\|_2^2 - 2\int_0^\infty \frac{e^{s/2}C_u(s) - \|u\|_2^2}{e^s - e^{-s}}\,\mathrm{d}s. \tag{2.8}$$

The subtraction remains coupled at zero. Using $\|u - \tau_s u\|_2^2 = 2(\|u\|_2^2 - C_u(s))$, this is $D_\Gamma(u) - c_\Gamma\|u\|_2^2$. Indeed the additional constant is

$$2\int_0^\infty \frac{e^{s/2} - 1}{e^s - e^{-s}}\,\mathrm{d}s = \psi(1/2) - \psi(1/4) = \frac{\pi}{2} + \log 2. \tag{2.9}$$

The digamma integral, reflection, and duplication formulas used in this calculation are [4, Eqs. 5.9.16, 5.5.4, 5.5.8]. This proves the precise normalization in (2.6). Compact smooth transforms decay faster than any power on $|\Im z| \le 1/2$. The unconditional bound $N_\zeta(T) = O(T\log(2+T))$ then proves absolute convergence. Polarization gives (2.7); conjugation invariance of the zero multiset makes the displayed pairing Hermitian. More generally, the mixed series is absolutely convergent whenever both transforms are $O((1+|z|)^{-2})$ uniformly on $|\Im z| \le 1/2$. □

The only arithmetic zero-counting input used in this paper is the classical asymptotic

$$N_\zeta(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + O(\log T), \tag{2.10}$$

counting multiplicities. For a precise stronger bound see [3, Corollary 1.2]; none of its explicit numerical constants is used. The unconditional strip $0 < \Re\rho < 1$ and its symmetries are recalled in [4, Section 25.10]. The factor $1/2$ in our definition of ξ does not affect its zeros.

2.3 Source cancellation and cutoff continuity

Lemma 2.4. *On even functions with finite*

$$M(u) = \|u\|_{H^1(\mathbb{R})} + \|e^{|x|}u\|_2, \tag{2.11}$$

one has $|Q(u,v)| \le CM(u)M(v)$. For every fixed $j \ge 0$ and every compact smooth even k,

$$Q(\kappa^{(2j)}, k) = 0. \tag{2.12}$$

Proof. The Gamma estimate follows by using $\|u - \tau_s u\|_2 \le s\|u'\|_2$ for $s \le 1$ and $\|u - \tau_s u\|_2 \le 2\|u\|_2$ for $s \ge 1$. The pole is bounded in M by Cauchy–Schwarz. For the primes,

$$|\langle u, \tau_s v\rangle_2| \le e^{-|s|}\|e^{|x|}u\|_2\|e^{|x|}v\|_2, \qquad \sum_{n\ge 2}\Lambda(n)n^{-3/2} < \infty. \tag{2.13}$$

This proves the asserted continuity without a sign assumption.

To pass the explicit formula to the source, also use

$$S_2(u) = \sum_{r=0}^{2}\int_{\mathbb{R}} e^{|x|}|u^{(r)}(x)|\,\mathrm{d}x.$$

Integration by parts gives $|F_u(z)| \le CS_2(u)(1+|z|)^{-2}$ on $|\Im z| \le 1/2$. Consequently the polarized zero sum is jointly continuous under S_2 convergence, since $\sum_\rho (1+|z_\rho|)^{-4} < \infty$ by (2.10). Choose even smooth cutoffs $\chi_R(x) = \chi(x/R)$, with $\chi = 1$ on $[-1,1]$ and supported in $[-2,2]$. For $u = \kappa^{(2j)}$, theta decay and the product rule imply

$$M(\chi_R u - u) \longrightarrow 0, \qquad S_2(\chi_R u - u) \longrightarrow 0.$$

Thus the polarized identity (2.7) passes to (u,k). Since $F_u(z) = (-1)^j z^{2j}\Xi(z)$ vanishes at every zero argument, the mixed pairing is zero. This proves (2.12). □

Proposition 2.5 (Exact source transform)**.** *On $\mathcal{C}$ one has*

$$Q(\kappa h) = \mathcal{E}(h) - \frac{1}{2}\operatorname{Var}_\nu(h). \tag{2.14}$$

On the minimum domain $\mathcal{D}$,

$$\begin{aligned}\mathcal{E}(h) = \frac{1}{2}\|h\|_{\mathcal{H}}^2 + D_\Gamma(\kappa h) - c_\Gamma\|\kappa h\|_2^2 \\ - 2\sum_{n\ge 2} c_n \Re\langle \kappa h, \tau_{s_n}(\kappa h)\rangle_2.\end{aligned} \tag{2.15}$$

Proof. For real a, b and complex u, v the edge identity is

$$|au - bv|^2 - (a-b)(a|u|^2 - b|v|^2) = ab|u - v|^2. \tag{2.16}$$

Subtract $Q(\kappa, \kappa|h|^2) = 0$ from $Q(\kappa h)$, using Lemma 2.4. For the continuous part apply (2.16); for the primes symmetrize both orientations. The resulting coefficient of the positive prime-edge integral is exactly c_n, not $2c_n$. The local c_Γ terms cancel. Evenness and (2.1) give

$$\mathfrak{m}(\kappa h) = \frac{1}{2}\int h \,\mathrm{d}\nu, \qquad \mathfrak{m}(\kappa|h|^2) = \frac{1}{2}\|h\|_{\mathcal{H}}^2, \qquad \mathfrak{m}(\kappa) = \frac{1}{2}.$$

Hence the pole difference is $-\operatorname{Var}_\nu(h)/2$, proving (2.14). Shift truncation, the coupled differences, and the continuity bounds above justify the expansion. Substitution of (2.5) gives (2.15) on the core.

For the extension let $Uh = \sqrt{2\kappa c}\, h$ and $b = \sqrt{\kappa/(2c)}$, so that $\kappa h = bUh$. The map U is unitary, whereas multiplication by b is merely bounded. Theta decay gives $b(x) \le Ce^{-|x|}$, and therefore

$$K_p = \sum_{n\ge 2} c_n M_b(\tau_{s_n} + \tau_{-s_n}) M_b \tag{2.17}$$

converges absolutely in operator norm: $\|M_b \tau_s M_b\| \le C^2 e^{-|s|}$ and (2.13) supplies summability. On the core the prime term is $\langle Uh, K_p Uh\rangle_2$. The form D_Γ is closed by Plancherel, with nonnegative multiplier $2\int_0^\infty \rho(s)(1-\cos(ts))\,\mathrm{d}s$.

These bounds also give a second proof of closability directly from the *core* identity (2.15). If $h_k \in \mathcal{C}$ tends to zero in $\mathcal{H}$ and is $\mathcal{E}$-Cauchy, then for $d = h_k - h_\ell$ the bounded lower-order terms give

$$D_\Gamma(\kappa d) \le \mathcal{E}(d) + C\|d\|_{\mathcal{H}}^2.$$

Thus $\kappa h_k = bUh_k$ is Cauchy in the D_Γ graph norm and has L^2 limit zero. Closedness yields $D_\Gamma(\kappa h_k) \to 0$; the core identity and the same bounded terms yield $\mathcal{E}(h_k) \to 0$. This argument uses no prior extension of the identity to $\mathcal{D}$.

Applied to differences of form-Cauchy core approximants to an arbitrary $h \in \mathcal{D}$, the same estimate gives convergence of their physical images in the D_Γ graph norm. All terms thus pass to the $\mathcal{H}$ limit. This proves the identity on the specified minimum closure. □

Lemma 2.6 (A minimum-domain admission criterion)**.** *Suppose h is smooth and even, $h \in \mathcal{H}$, and $u = \kappa h$ and u' decrease faster than every exponential. Then $h \in \mathcal{D}$ and (2.15) applies to h.*

Proof. For the cutoffs used above, $\chi_R h \to h$ in $\mathcal{H}$ and $\chi_R u \to u$ in M. For $d = (\chi_R - \chi_S)h \in \mathcal{C}$, the *compact* identity gives

$$0 \le \mathcal{E}(d) = Q(\kappa d) + \frac{1}{2}\operatorname{Var}_\nu(d) \le CM(\kappa d)^2 + \frac{1}{2}\|d\|_{\mathcal{H}}^2 \longrightarrow 0. \tag{2.18}$$

The compact approximants are Cauchy in (1.5); their $\mathcal{H}$ limit therefore belongs to the minimum domain. This criterion does not replace minimum admission by finite maximal energy. □

Proposition 2.7 (The known source space)**.** *For every fixed $j \ge 0$, $f_j \in \mathrm{D}(L)$, $Lf_j = 0$, and $\langle 1, f_j\rangle_{\mathcal{H}} = 4^{-j}$. Consequently $\mathcal{N} \subset \ker L$.*

Proof. Equation (2.3) gives $|f_j(x)| \le C_j e^{4j|x|}$, so $f_j \in \mathcal{H}$. Lemma 2.6 applies because $\kappa f_j = \kappa^{(2j)}$. Polarizing (2.14) and using cutoffs and (2.12), the form of L satisfies $\ell(f_j, g) = 0$ for every core g. Continuity in the form norm extends this to all $g \in \mathcal{D}$. The closed-form representation theorem gives $f_j \in \mathrm{D}(L)$ and $Lf_j = 0$. Integration by parts yields

$$\langle 1, f_j\rangle_{\mathcal{H}} = 2\int \kappa^{(2j)}(x)c(x)\,\mathrm{d}x = 4^{-j}2\int \kappa(x)c(x)\,\mathrm{d}x = 4^{-j}.$$

The kernel of the closed operator L is $\mathcal{H}$-closed, so it contains the closed span (1.7). This proves an inclusion, not equality. □

3 Exact constraints and clustered zero modes

Let Z be the zero multiset of Ξ. For a zero z of multiplicity d and $0 \le r < d$, define

$$v_{z,r}(x) = \frac{\partial_z^r \cos(zx)}{c(x)}. \tag{3.1}$$

Lemma 3.1. *Every $v_{z,r}$ belongs to $\mathcal{D} \cap \mathcal{N}^\perp$. Its real and imaginary parts have the same membership.*

Proof. The functions are even, smooth, and have at most polynomial times exponential growth. Theta decay gives their $\mathcal{H}$ membership and the hypotheses of Lemma 2.6. For every $j \ge 0$,

$$\begin{aligned} \langle f_j, v_{z,r} \rangle_{\mathcal{H}} &= 2\partial_z^r \int_{\mathbb{R}} \kappa^{(2j)}(x) \cos(zx)\, \mathrm{d}x \\ &= 2\partial_z^r \left[(-1)^j z^{2j} \Xi(z) \right] = 0. \end{aligned} \tag{3.2}$$

All differentiations and integrations by parts are justified by theta decay, for each fixed order. The last equality uses $r < d$. Thus every generator moment, including the mean, vanishes exactly. Closed span gives $\mathcal{N}^\perp$; the generators are real, so real and imaginary parts qualify. □

Lemma 3.2 (Clusters from zero counting)**.** *For every sufficiently large integer m, there are $m+1$ zero arguments $z_0, \dots, z_m$, counted with multiplicity, and a real λ_m such that*

$$\begin{aligned} T_m = e^{m^3}, \qquad & T_m \le \lambda_m \le 2T_m, \\ \lambda_m \le \Re z_j \le \lambda_m + 1/m, \qquad & |\Im z_j| < 1/2. \end{aligned} \tag{3.3}$$

Proof. Equation (2.10) gives $N_\zeta(2T_m) - N_\zeta(T_m) \sim T_m m^3/(2\pi)$. Partition $[T_m, 2T_m]$ into at most $mT_m + 1$ intervals of length at most $1/m$. The average number of zeros is asymptotic to $m^2/(2\pi)$, which exceeds m for large m. Choose an interval containing at least $m+1$ zeros and let λ_m be its left endpoint. The classical strip gives the imaginary-part bound. An endpoint convention does not affect the argument. Repeated nodes are taken no more often than their multiplicities. □

No zero at height T_m is computed. The choice e^{m^3} keeps $\log \lambda_m$ polynomial in m while making λ_m dominate the oscillatory error below.

Write $[z_0, \dots, z_m]F$ for the divided difference, with its confluent interpretation at repeated nodes, and define

$$P_m(x) = m! i^{-m} [z_0, \dots, z_m] \cos(zx), \qquad p_m = P_m / c. \tag{3.4}$$

This is a finite linear combination of the modes in (3.1); therefore $p_m \in \mathcal{D} \cap \mathcal{N}^\perp$ exactly. For distinct nodes, for example,

$$P_m(x) = m! i^{-m} \sum_{j=0}^{m} \frac{\cos(z_j x)}{\prod_{k \ne j} (z_j - z_k)}.$$

Estimating these individual coefficients would be inappropriate when nodes are close. We instead use the simplex representation.

Let θ have uniform probability distribution on the simplex $\theta_j \ge 0$, $\sum_{j=0}^m \theta_j = 1$, and put

$$A_m(x) = \mathbb{E}_\theta \exp\left(ix \sum_{j=0}^{m} \theta_j (z_j - \lambda_m) \right). \tag{3.5}$$

The classical divided-difference integral formula, discussed in [2], is

$$[z_0, \dots, z_m]F = \frac{1}{m!}\mathbb{E}_\theta F^{(m)}\left(\sum_j \theta_j z_j\right). \tag{3.6}$$

It holds for entire F and complex nodes: repeated integration along complex straight segments proves the formula for distinct nodes, and continuous extension gives the confluent case. Equivalently, its power series proof follows from the simplex moments and complete homogeneous polynomials. Applying it to the two exponentials in the cosine gives

$$P_m(x) = \frac{x^m}{2}\left(e^{i\lambda_m x}A_m(x) + (-1)^m e^{-i\lambda_m x}A_m(-x)\right). \tag{3.7}$$

Lemma 3.3. *For $m \geq 2$ and real x,*

$$\begin{aligned} |A_m(\pm x)|,\ |A_m'(\pm x)| &\leq e^{|x|/2}, \\ |P_m(x)| &\leq |x|^m e^{|x|/2}, \\ |P_m'(x)| &\leq \left(m|x|^{m-1} + (\lambda_m + 1)|x|^m\right)e^{|x|/2}. \end{aligned} \tag{3.8}$$

If $0 \leq x \leq b_m := m/4$, then

$$\Re A_m(\pm x) \geq \cos(1/4)e^{-x/2}. \tag{3.9}$$

Proof. Each convex combination of $z_j - \lambda_m$ has real part in $[0, 1/m]$ and imaginary part in $(-1/2, 1/2)$; its modulus is less than one for $m \geq 2$. This proves the first line and its derivative bound directly from (3.5), and the next two lines follow from (3.7). On the indicated interval the phase magnitude is at most $x/m \leq 1/4$, whereas the positive exponential amplitude is at least $e^{-x/2}$. This proves (3.9). □

4 Normalization and separation of theta moments

Put $W = \kappa/c$ on $[0, \infty)$ and define

$$d_m = \int_0^{b_m} x^{2m}W(x)e^{-x}\,\mathrm{d}x. \tag{4.1}$$

This is a concrete positive source moment for the constructed functions, not an unknown optimal constant for the critical form inequality.

Lemma 4.1 (Control of the oscillatory cross term)**.** *For all sufficiently large m,*

$$\|p_m\|_{\mathcal{H}}^2 \geq d_m/2. \tag{4.2}$$

Proof. Expand (3.7). The diagonal terms and (3.9) give

$$\begin{aligned} \int_0^{b_m} W|P_m|^2 &\geq \frac{\cos^2(1/4)}{2}d_m - \frac{1}{2}|I_m|, \\ I_m &= \int_0^{b_m} x^{2m}W(x)A_m(x)\overline{A_m(-x)}e^{2i\lambda_m x}\,\mathrm{d}x. \end{aligned} \tag{4.3}$$

Let $B_m(x) = x^{2m}W(x)A_m(x)\overline{A_m(-x)}$. Then $B_m(0) = 0$, and integration by parts yields

$$|I_m| \leq \frac{|B_m(b_m)| + \int_0^{b_m}|B_m'|\,\mathrm{d}x}{2\lambda_m} \leq \frac{C(m+1)(1+m)^{2m}}{\lambda_m}. \tag{4.4}$$

Indeed (3.8) bounds the integrand in the numerator by

$$2mx^{2m-1}W(x)e^x + x^{2m}(|W'(x)| + 2W(x))e^x.$$

The fixed source weights have finite integrals and suprema by Lemma 2.1. For $m \geq 8$, $d_m \geq d_0 := \int_1^2 W(x)e^{-x}\,\mathrm{d}x > 0$. Since $\lambda_m \geq e^{m^3}$, the right side of (4.4) is $o(d_m)$. Finally, evenness and the exact weight give

$$\|p_m\|_{\mathcal{H}}^2 = 4\int_0^\infty W(x)|P_m(x)|^2\,\mathrm{d}x.$$

Together with (4.3) this proves (4.2). □

For a fixed $a \geq 0$ write

$$U_m(a) = \int_0^\infty x^{2m}\frac{\kappa(x)^2}{c(x)^2}e^{ax}\,\mathrm{d}x, \qquad M_m = \frac{\log m}{4}, \qquad C_0 = \pi e^4. \tag{4.5}$$

Lemma 4.2 (Separation of the two theta weights)**.** *For every fixed $a \geq 0$ there is $C_a < \infty$ such that, for all sufficiently large m,*

$$\frac{U_m(a)}{d_m} \leq C_a e^{-\sqrt{m}}, \qquad \frac{1}{d_m} \leq Ce^{-\sqrt{m}}. \tag{4.6}$$

The starting index may depend on a.

Proof. For large m, $[M_m + 1, M_m + 2] \subset [0, b_m]$. Since $e^{2M_m} = \sqrt{m}$ and $c(x) \leq e^{x/2}$, the lower theta estimate gives

$$d_m \geq 2\pi^2(M_m + 1)^{2m}e^{-C_0\sqrt{m}}. \tag{4.7}$$

In particular $d_m^{-1} = O(e^{-\sqrt{m}})$.

Split $U_m(a)$ at M_m and b_m. On $[0, M_m]$ it is bounded by $C_a M_m^{2m}$, since its fixed source weight is integrable. Consequently

$$\frac{U_{m,\mathrm{low}}(a)}{d_m} \leq C_a \exp\left(-\frac{2m}{M_m + 1} + C_0\sqrt{m}\right). \tag{4.8}$$

Here $\log(M/(M+1)) \leq -1/(M+1)$, and $m/\log m$ dominates $\sqrt{m}$, giving the required rate for this part.

On $[M_m, b_m]$ the ratio of the numerator weight to that in d_m is

$$\frac{\kappa(x)}{c(x)}e^{(a+1)x} \leq C_a e^{-(\pi/2)e^{2x}} \leq C_a e^{-(\pi/2)\sqrt{m}}. \tag{4.9}$$

This follows from (2.2) by absorbing a fixed exponential factor into half of the theta exponent. Integrating bounds the middle contribution by the last quantity times d_m.

On $[b_m, \infty)$ the source weight is at most $C_a e^{-\pi e^{2x}}$. For $x \geq m/4$ and all sufficiently large m, $2m\log x \leq (\pi/2)e^{2x}$ uniformly. It suffices to check the ratio at $m/4$ and use the subsequent decrease of $\log x/e^{2x}$. The tail is therefore bounded by $C_a e^{-ce^{m/2}}$ for a fixed $c > 0$. Divide by $d_m \geq d_0 > 0$ and combine the three parts to obtain (4.6). □

Splitting at $x = 1$ also gives

$$U_{m-1}(1) \leq U_m(1) + C. \tag{4.10}$$

Thus any fixed polynomial in m times $(U_m(a) + U_{m-1}(1) + 1)/d_m$ tends to zero. The estimates are asymptotic analytic inequalities, not finite numerical enclosures.

5 Energy control and escape from the center

Let $u_m = \kappa p_m = \kappa P_m/c$. Lemma 3.3 and $|(\kappa/c)'| \le Ce^{2x}\kappa/c$ for $x \ge 0$ imply

$$\begin{aligned} \|u_m\|_2^2 &\le 2U_m(1), \\ \|e^{|x|}u_m\|_2^2 &\le 2U_m(3), \\ \|u_m'\|_2^2 &\le C((\lambda_m+1)^2U_m(1) + U_m(5) + m^2U_{m-1}(1)). \end{aligned} \tag{5.1}$$

These constants are independent of the chosen cluster and of m.

Lemma 5.1 (Logarithmic Gamma estimate)**.** *For $u \in H^1(\mathbb{R})$ and $\Omega \ge 2$,*

$$D_\Gamma(u) \le C\left(\Omega^{-2}\|u'\|_2^2 + (1+\log\Omega)\|u\|_2^2\right). \tag{5.2}$$

Proof. Split the shift integral at $1/\Omega$ and 1. On $(0, 1/\Omega)$ use $\|u-\tau_s u\|_2 \le s\|u'\|_2$ and $\rho(s) \le C/s$. On $(1/\Omega, 1)$ use $2\|u\|_2$, giving $C\log\Omega$. On $(1,\infty)$ the kernel is integrable. This proves the bound directly from the shift integral, without a separate multiplier asymptotic. □

With $\Omega = \lambda_m + 2$ one has $\log\Omega \le m^3 + C$. Equations (4.6), (4.10), and (5.1) give

$$D_\Gamma(u_m) + \|e^{|x|}u_m\|_2^2 \le Cm^3e^{-\sqrt{m}}d_m. \tag{5.3}$$

The prime term retains its arithmetic weights:

$$2\sum_{n\ge2} c_n|\langle u_m, \tau_{s_n}u_m\rangle_2| \le 2\sum_{n\ge2}\frac{\Lambda(n)}{n^{3/2}}\|e^{|x|}u_m\|_2^2. \tag{5.4}$$

Using the exact identity (2.15), we obtain

$$|\mathcal{E}(p_m) - \tfrac12\|p_m\|_{\mathcal{H}}^2| \le Cm^3e^{-\sqrt{m}}d_m. \tag{5.5}$$

Choose r_m to be the real or imaginary part of p_m having the larger $\mathcal{H}$ norm, and put

$$h_m = \frac{r_m}{\|r_m\|_{\mathcal{H}}}, \qquad \|r_m\|_{\mathcal{H}}^2 \ge \frac12\|p_m\|_{\mathcal{H}}^2 \ge \frac14 d_m. \tag{5.6}$$

These are real smooth even members of $\mathcal{D}\cap\mathcal{N}^\perp$. The estimates (5.1)–(5.4) hold for each real or imaginary part with the same upper bounds. Applying the source identity to the chosen part separately proves (1.8) after division by (5.6). This step does not infer a signed bound for one part from possible cancellation between the two signed errors.

For fixed $T > 0$, (3.8), (4.7), and (5.6) imply

$$\|\mathbf{1}_{[-T,T]}h_m\|_{\mathcal{H}}^2 \le C_T\left(\frac{T}{M_m+1}\right)^{2m}e^{C_0\sqrt{m}} \longrightarrow 0. \tag{5.7}$$

The norm-one bound and compact truncation of arbitrary $\mathcal{H}$ tests then give weak convergence to zero. The same calculation with $T = M_m$ uses $\int_0^\infty W(x)e^x\,dx < \infty$ and yields the uniform moving-cutoff bound

$$\|\mathbf{1}_{[-M_m,M_m]}h_m\|_{\mathcal{H}}^2 \le C\exp\left(-\frac{2m}{M_m+1} + C_0\sqrt{m}\right). \tag{5.8}$$

This is $O(e^{-\sqrt{m}})$ after increasing the starting index. It completes the proof of (1.8)–(1.9).

6 A two-shift envelope and its minimum-domain transform

6.1 Safe shifts and existence of a positive envelope

For $|t| < a_* := (\log 2)/2$ define

$$\Phi_t(x) = \frac{\kappa(x+t) + \kappa(x-t)}{2\kappa(x)}, \qquad q_t = \Phi_t - \cosh(t/2). \tag{6.1}$$

Lemma 6.1. *For every safe real shift, $q_t \in \mathcal{N} \cap \mathrm{D}(A)$, $\langle 1, q_t\rangle_{\mathcal{H}} = 0$, and $Aq_t = q_t/2$.*

Proof. The theta series is holomorphic for $|\Im x| < \pi/4$. On compact subsets of the connected region

$$\mathcal{U} = \left\{ t \in \mathbb{C} : |\Im t| < \pi/4,\ e^{-2|\Re t|}\cos(2\Im t) > 1/2 \right\}, \tag{6.2}$$

the shifted source, squared and divided by κ, is bounded in the tails by a fixed exponential times

$$\exp\left(-\pi\left[2e^{-2|\Re t|}\cos(2\Im t) - 1\right]e^{2|x|}\right).$$

The bracket has a positive lower bound there. Local domination, also for local t derivatives, proves that q_t is $\mathcal{H}$-holomorphic. Near zero its Taylor series is

$$q_t = \sum_{j\geq 1} \frac{t^{2j}}{(2j)!}(f_j - 4^{-j}).$$

The quotient map to $\mathcal{H}/\mathcal{N}$ is therefore zero near zero, and hence on $\mathcal{U}$ by analytic continuation. This proves $q_t \in \mathcal{N}$ on the whole safe real interval without assuming a global Taylor radius. Changing variables gives $\int \Phi_t \,\mathrm{d}\nu = \cosh(t/2)$. Proposition 2.7 now gives the action and domain claims. □

Definition 6.2. An admissible envelope has parameters satisfying

$$0 < t_2 < t_1 < a_*, \qquad 2e^{-2t_2} > 1 + e^{-2t_1}, \qquad w_1, w_2 > 0, \tag{6.3}$$

such that q_{t_1}, q_{t_2} have no common real zero. Set

$$R = \sqrt{w_1 q_{t_1}^2 + w_2 q_{t_2}^2}, \qquad \Delta = R - \sqrt{w_1}\, q_{t_1}, \qquad K = \kappa R. \tag{6.4}$$

Lemma 6.3. *For each fixed $t_1 \in (0, a_*)$ an admissible t_2 exists. For any admissible parameters, $R > 0$, $\inf R > 0$, and R tends to infinity at both ends. The function $\Delta \geq 0$ and all its fixed derivatives decrease faster than every exponential. Moreover*

$$1 < \gamma := \sup_x \frac{\Delta(x)}{R(x)} \leq 2. \tag{6.5}$$

Proof. For fixed safe $t > 0$, the leading theta term gives

$$q_t(x) \sim \frac{1}{2} e^{-9t/2} \exp\left(\pi(1 - e^{-2t})e^{2|x|}\right). \tag{6.6}$$

Thus q_{t_1} is positive in both tails and, being nonzero and real analytic, has finitely many real zeros. Write $a_i = e^{-2t_i}$ and $t_* = -\frac{1}{2}\log((1+a_1)/2)$, so $0 < t_* < t_1$. At each zero x of q_{t_1}, the real analytic function $t \mapsto q_t(x)$ is not identically zero: as $t \to +\infty$ its shifted numerator tends to zero while $\cosh(t/2)$ diverges. Its zeros on $[0, t_*]$ are finite. Choose $t_2 \in (0, t_*)$ outside the finite union of these exceptions. This proves existence, without certifying a numerical pair.

In the tails $q_{t_1} > 0$ and $q_{t_2}/q_{t_1} \to 0$, giving

$$R \sim \sqrt{w_1} q_{t_1}, \qquad \Delta \sim \frac{w_2}{2\sqrt{w_1}} \frac{q_{t_2}^2}{q_{t_1}}. \tag{6.7}$$

The latter follows from $\Delta = w_2 q_{t_2}^2/(R + \sqrt{w_1} q_{t_1})$ *in the tails*; this quotient need not have a nonzero denominator everywhere. Its exponential coefficient is $-\eta$, where $\eta = \pi(2a_2 - 1 - a_1) > 0$. Fixed differentiation of the theta series and the quotient in the tails therefore gives

$$|\Delta^{(r)}(x)| \le C_r e^{C_r|x|} e^{-\eta e^{2|x|}} \tag{6.8}$$

for large $|x|$, weakening η if necessary. To justify the quotient derivatives, use $R = \sqrt{w_1} q_{t_1}(1 + (w_2/w_1)(q_{t_2}/q_{t_1})^2)^{1/2}$ there; fixed logarithmic derivative bounds grow only polynomially in $e^{2|x|}$. On the remaining compact interval R has no zero and is smooth. This proves all the asserted regularity and the positive infimum of R.

Finally $\Delta/R = 1 - \sqrt{w_1} q_{t_1}/R \in [0, 2]$. The nonzero mean-zero function q_{t_1} must be negative somewhere, because it is positive in the tails. At such a point $\Delta/R > 1$, proving (6.5). □

6.2 The bounded potential and the ground-state identity

Lemma 6.4 (A prime-weighted row bound)**.** *If $J \ge 0$ is continuous and $J(y) \le C_B e^{-B|y|}$ for some $B > 1/2$, then*

$$\sum_{n \ge 2} \frac{\Lambda(n)}{\sqrt{n}} (J(x + s_n) + J(x - s_n)) \le C_J c(x). \tag{6.9}$$

Proof. For positive integer r, the prime-adic valuations of the central binomial coefficient give $\psi(2r) - \psi(r) \le \log \binom{2r}{r} \le 2r \log 2$. Dyadic summation and monotonicity imply the elementary Chebyshev estimate $\psi(X) \le CX$. Thus

$$\sum_{k \le \log n < k+1} \frac{\Lambda(n)}{\sqrt{n}} \le C e^{k/2}.$$

Multiply by the supremum of $J(x \pm s)$ on each unit interval. The inequality $k \le |x| + |x \pm k|$ bounds the result by $Ce^{|x|/2}$ times a uniformly summable lattice series with decay exponent $B - 1/2$. Since $e^{|x|/2} \le 2c(x)$, this proves the claim. The weights have not been discarded. □

Proposition 6.5 (Minimum-domain ground-state transform)**.** *For an admissible envelope, $\Delta, R \in \mathrm{D}(A)$ and $A\Delta$ is bounded. The function*

$$V = \frac{1}{2} - \frac{AR}{R} = \frac{\Delta/2 - A\Delta}{R} \tag{6.10}$$

is bounded and, for some $b_R > 0$,

$$|V(x)| \le C_R e^{-b_R e^{2|x|}} \quad \text{for large } |x|. \tag{6.11}$$

For every $h \in \mathcal{D}$ the following transformed energy is finite and satisfies

$$\begin{aligned} D_R(h) &= \iint K(x)K(z) \Big| \frac{h(x)}{R(x)} - \frac{h(z)}{R(z)} \Big|^2 \, \mathrm{d}\mu(x - z) \, \mathrm{d}x, \\ \mathcal{E}(h) - \frac{1}{2}\|h\|_{\mathcal{H}}^2 &= D_R(h) - \int_{\mathbb{R}} V|h|^2 \, \mathrm{d}\nu. \end{aligned} \tag{6.12}$$

Proof. Equation (6.8) and Lemma 2.6 give $\Delta \in \mathcal{D}$. Define the coupled expression

$$B_\Delta(x) = \frac{1}{c(x)} \int \kappa(z)(\Delta(x) - \Delta(z)) \, \mathrm{d}\mu(x - z). \tag{6.13}$$

For shifts of size less than one use the bounded derivative of Δ and $\int_0^1 s\rho(s) \, \mathrm{d}s < \infty$; do not separate the two singular convolutions. The large continuous shifts have finite mass. For the primes use Lemma 6.4 with $J = \kappa$, and the boundedness of Δ. These estimates prove $\|B_\Delta\|_\infty < \infty$. Symmetrization of the half measure against a compact test yields $\mathcal{E}(\Delta, \varphi) = \langle B_\Delta, \varphi \rangle_{\mathcal{H}}$. Density in $\mathcal{D}$ and the representation theorem give $\Delta \in \mathrm{D}(A)$ and $A\Delta = B_\Delta$.

Lemma 6.1 and $R = \sqrt{w_1} q_{t_1} + \Delta$ now give $R \in \mathrm{D}(A)$ and $AR = \sqrt{w_1} q_{t_1}/2 + A\Delta$. This proves (6.10). The positive infimum and tail asymptotic of R, together with the bounded numerator in (6.10), prove (6.11) and boundedness. No global bound for R' is required.

For $h \in \mathcal{C}$, $g = h/R$ and $|h|^2/R$ are compact smooth functions. Applying (2.16) with $a = R(x)$ and $b = R(z)$ gives

$$D_R(h) = \mathcal{E}(h) - \mathcal{E}(R, |h|^2/R) = \mathcal{E}(h) - \int \frac{AR}{R} |h|^2 \, \mathrm{d}\nu.$$

Shift truncation, edge Cauchy–Schwarz for the mixed term, and monotone convergence for the positive term justify the integration. This is (6.12) on the core.

For $h \in \mathcal{D}$, let core functions $h_k \to h$ in the form norm. Boundedness of V and the identity applied to differences show that h_k/R is Cauchy in the positive transformed edge energy. It also converges to h/R in $L^2(R^2 \, \mathrm{d}\nu)$. A subsequence converges at both endpoints almost everywhere for the sigma-finite transformed edge measure, because its restricted endpoint marginals are Lebesgue absolutely continuous, including every countable atomic translation. The edge-L^2 limit is thus the actual difference of h/R. Passing to the limit proves both finiteness and the identity on the minimum domain. No maximal-domain equality has been used. □

6.3 Completion of the escape theorem and its boundary

Split the potential integral at $M_m = (\log m)/4$. Equations (5.8) and (6.11) imply, for some $a_R > 0$,

$$\int |V||h_m|^2 \, \mathrm{d}\nu \le \|V\|_\infty \|\mathbf{1}_{[-M_m, M_m]} h_m\|_{\mathcal{H}}^2 + \sup_{|x| > M_m} |V(x)| \le C_R e^{-a_R \sqrt{m}}. \tag{6.14}$$

The ground-state identity and (1.8) prove (1.10), after decreasing a_R if necessary. This completes Theorem 1.1. Substituting this sequence proves Corollary 1.2.

For real $h = Rg \in \mathcal{D} \cap \mathcal{N}^\perp$, put

$$F(g) = \int g^2 R^2 V \, \mathrm{d}\nu, \qquad D(g) = D_R(Rg). \tag{6.15}$$

The desired critical comparison is $F(g) \le D(g)$. Since $\mathcal{N} \subset \ker L$, the space $\mathcal{N}$ reduces L, and decomposing a form vector into its $\mathcal{N}$ and $\mathcal{N}^\perp$ parts shows that this comparison is equivalent to $L \ge 0$. This is an equivalence, not an additional estimate. On the constructed sequence both $F(h_m/R)$ and $D(h_m/R)$ tend to zero; their relative ordering is not determined.

7 Scope, verification, and data availability

The functions in Theorem 1.1 are noncompact. They satisfy all conditions from $\mathcal{N}$ exactly and belong to the prescribed minimum form domain. They are not compact tail bumps with only finitely many moments removed. Their small mass on every fixed center does not assert exact

vanishing on such a center. The bound is stated for sufficiently large m; neither a finite starting index nor numerical values of its constants have been certified. The cluster height e^{m^3} is an analytic existence device, not a performed or proposed zero computation.

The critical sign in (6.15) and the Riemann hypothesis remain open in this work. We do not prove $\mathcal{N} = \ker L$, global completeness of the zero modes in $\mathcal{N}^\perp$, membership of the entire sequence in $\mathrm{D}(L)$, or $\|Lh_m\| \to 0$. In particular, convergence of form Rayleigh quotients is not presented as an operator Weyl-sequence argument or an essential-spectrum conclusion.

The accompanying research artifacts include an exact integer-coefficient check of (2.16) and of the Young square used in Appendix B. These checks verify polynomial identities only; they do not verify the explicit formula, analytic estimates, domain arguments, or zero counting. No theta, zero, physical integral, FFT, Gram, or eigenvalue computation is used to establish these results. Compilation and rendering checks concern document integrity and layout.

The proof notes, source and dependency audits, reproduction instructions, and saved exact algebra records are preserved at the following fixed repository commit:

> https://github.com/serabing-hash/theta-weighted-certificate/commit/2447014b009ed432b353d6a3817294df59ff712e.

This artifact snapshot was publicly accessible before the manuscript's repository version was created. Its file hashes identify the supporting material; they are not mathematical proof certificates.

A Local density and compact-support uniqueness

Proposition A.1. *For each finite $S > 0$, the restrictions of the modes (3.1) have dense linear span in $L^2_{\mathrm{even}}([-S, S], \nu; \mathbb{C})$.*

Proof. If v annihilates all restricted modes, then

$$B(z) = 2 \int_{-S}^{S} \kappa(x) \overline{v(x)} \cos(zx)\, \mathrm{d}x$$

is entire and $|B(z)| \le C_v e^{S|\Im z|}$. By orthogonality it vanishes at every zero of Ξ to at least its multiplicity. A nonzero entire function of this exponential type has $O(r)$ zeros in $|z| \le r$, counting multiplicity: remove any finite-order zero at zero and apply Jensen's formula at $2r$. Equation (2.10) instead supplies a superlinear count. Thus $B \equiv 0$, and Fourier uniqueness for the compact L^1 function $\kappa\overline{v}$ gives $v = 0$. This proves density. Real and imaginary parts give the real version. □

Proposition A.2. *If $h \in \mathcal{N}^\perp$ has compact support, then $h = 0$.*

Proof. The even function $f = hc$ is compact L^1, and $\kappa * f(t)$ is holomorphic on $|\Im t| < \pi/4$ by the theta series and compact support. At zero, all even derivatives vanish by the generator moments; all odd derivatives vanish by parity. The identity theorem gives $\kappa * f = 0$ on the strip. Fourier transformation on the real line yields $\Xi \widehat{f} = 0$. The real zeros of the nonzero entire Ξ are isolated, so continuity of $\widehat{f}$ and Fourier uniqueness give $f = 0$. □

This argument does not apply to arbitrary unbounded tails, since hc need not be tempered for general $h \in \mathcal{H}$. Proposition A.1 is only a local restriction statement and does not imply a global expansion of $\mathcal{N}^\perp$ in zero modes.

B An oriented signed-jump estimate

For real $h \in \mathcal{C}$ and $g = h/R$, define

$$J(g) = \iint \kappa(x)\kappa(z)(\Delta(z) - \Delta(x))(R(x)g(x)^2 - R(z)g(z)^2)\,\mathrm{d}\mu(x-z)\,\mathrm{d}x. \tag{B.1}$$

Let $\Omega_+ = \{(x,z) : \Delta(z) > \Delta(x)\}$. The integrand is symmetric under swapping endpoints, so J is twice its integral over Ω_+. For $\alpha > 0$ Young's inequality gives

$$R(x)g(x)^2 - R(z)g(z)^2 \le (1+\alpha)R(x)(g(x)-g(z))^2 + ((1+1/\alpha)R(x) - R(z))g(z)^2.$$

On Ω_+, $\Delta(z) - \Delta(x) \le \Delta(z) \le \gamma R(z)$. Twice the integral of the symmetric transformed squared difference over Ω_+ is at most its full integral, including possible equality edges. Consequently

$$\begin{aligned} J(g) &\le (1+\alpha)\gamma D(g) + \int_{\mathbb{R}} g(z)^2 \widetilde{W}(z)\,\mathrm{d}z, \\ \widetilde{W}(z) &= 2\kappa(z) \int_{\Delta(z)>\Delta(x)} \kappa(x)(\Delta(z)-\Delta(x)) \\ &\qquad \cdot \left[(1+1/\alpha)R(x) - R(z)\right]_+ \mathrm{d}\mu(x-z). \end{aligned} \tag{B.2}$$

The last integral in the first line is with respect to Lebesgue measure. For the stated compact tests, the Δ difference removes the small-shift singularity; large shifts use the prime-weighted row bound for K. No unrestricted nonlinear form pairing on all of $\mathcal{D}$ is asserted.

This orientation saves a factor of two in the displayed jump coefficient, but (6.5) gives

$$1 + \alpha < (1+\alpha)\gamma \le 2(1+\alpha).$$

Moreover $F(g) = \frac{1}{2}\int \Delta R g^2\,\mathrm{d}\nu + J(g)$ on the core. Thus (B.2) does not absorb D with a coefficient below one and does not prove the critical comparison.

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